

Hussein Khalil

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EDUCATION

Purdue University, Department of Pharmacy Practice **West Lafayette, IN**
PhD Health Services, Outcomes, and Policy Research (GPA: 3.9/4) 2025
Dissertation: *Cumulative Anticholinergic Burden and Delirium In Critically Ill Patients*

Lebanese University, College of Pharmacy **Beirut, Lebanon**
M.S Pharmacoepidemiology and Clinical Pharmacy (GPA: 4.0/4.0) 2018
Thesis: *Attitudes and Perceptions of Stroke Signs and Risk Factors among Lebanese Older Adults*

Beirut Arab University, College of Pharmacy **Beirut, Lebanon**
B.S of Pharmacy 2017

EXPERIENCE

Boehringer Ingelheim **Ridgefield, CT**
HEOR Intern, Mental Health May 2024 – August 2024

- Conducted a targeted review of 52 investigational drugs for schizophrenia, culminating in a cross-functional presentation to key stakeholders and a conference abstract
- Built an economic burden model for schizophrenia using CMS, MEPS, and HCUP data, quantifying the direct and indirect cost burden by symptom clusters and support value and access strategy
- Led study protocol development for a multi-aim study to describe the schizophrenia patient population, treatment patterns, and outcomes using Medicaid T-MSIS data
- Selected and validated datasets for a geospatial analysis tool that identified psychiatric workforce shortages in areas with high schizophrenia burden, aligning results with value and access messaging

Takeda Pharmaceuticals **Ridgefield, CT**
Global Evidence and Outcomes Intern, Neuroscience May 2023 – August 2023

- Evaluated global real-world data sources across Europe and Asia to identify fit-for-use data for RWE studies across therapeutic areas. Results were synthesized into a decision guidance tool to inform portfolio evidence strategy.
- Led a multistage process including an in-depth literature review and targeted stakeholder interviews to guide the design and execution of indirect treatment comparisons for narcolepsy assets. Results were synthesized into a network meta-analysis manual and presented to GEO teams

Purdue University **Indianapolis, IN**
Postdoctoral Research Fellow, Allen Lab September 2025 – Current

- Leads grant writing on target trial emulation studies using SEER-Medicare
- Performs data analysis and preparation of scientific articles in oncology, clinical pharmacy, and health disparities research
- Co-mentors PharmD and graduate students engaged in research projects affiliated with the lab

Purdue University **West Lafayette, IN**
Graduate Research Assistant, Department of Pharmacy Practice January 2021 – May 2025

- Led two real-world analyses focused on long-term use of anticholinergics and delirium prevalence, severity, and mortality in critically ill patients using electronic health records data with sample sizes over 5,000 patients
- Identified major research gaps through a systematic review of 23 studies on community-based anticholinergic use and delirium
- Developed cumulative exposure algorithms for anticholinergic burden, addressing dose and duration using electronic health records medication order and dispensing data
- Co-developed and implemented a SAS-based pipeline that integrates structured and unstructured medication data to enhance completeness of medication order data ascertainment by up to 95% for a pragmatic clinical trial (MI-CARE) using a dataset of over 24,450 medication orders

Purdue University*Teaching Assistant, School of Pharmacy***West Lafayette, IN**

August 2020 – December 2021

- Assisted in teaching “Introduction to Patient Care” and “Patient Safety and Informatics”
- Taught and evaluated pharmacy practice labs for PharmD students for a year and a half
- Prepared and presented “Population Health Management” lectures for a class of 120+ PharmD students
- Co-mentored 4 PharmD students on research topics in geriatrics and pharmacoepidemiology, resulting in 5 student abstracts presented at annual conferences

American University of Beirut*Research Contractor, Hariri School of Nursing***Beirut, Lebanon**

October 2019 – November 2019

- Drafted and revised two manuscripts on implicit rationing of care for publication, integrating evidence with study findings and ensuring adherence to journal guidelines. Collaborated with investigators to refine research framing, supporting successful manuscript submission and dissemination.

Pharmacist in Charge*Ideal Pharmacy***Beirut, Lebanon**

June 2017– August 2020

- Oversaw all pharmacy operations including medication dispensing, inventory management, and compliance with Ministry of Health regulations

LEADERSHIP AND SERVICE**Purdue University***Graduate Student Council Representative, School of Pharmacy***West Lafayette, IN**

August 2023 – May 2025

- Represented pharmacy graduate students within Purdue’s College of Pharmacy council. Coordinated professional development activities, workshops, and symposia, including the biannual Tyler Symposium
- Advocated for student needs in curriculum and research support, strengthening academic engagement and resources

ISPOR Purdue Chapter*President,***West Lafayette, IN**

August 2022 – July 2023

Treasurer,

August 2021 – July 2022

- Directed educational and networking sessions for PharmD and graduate students interested in Health Economics and Outcomes Research
- Expanded chapter visibility by initiating social and academic programming that engaged over 30 members

AWARDS

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| • Ross-Lynn Research Scholar Award | 2023 |
| • Best Poster Award – Purdue College of Pharmacy Research Day | 2022 |
| • Teaching Academy Award – Purdue College of Pharmacy Nominee | 2022 |
| • Keiny Teaching Award – Excellence in teaching, Purdue College of Pharmacy | 2021 |

TECHNICAL AND ANALYTICAL SKILLS**Statistical Programming:** R, SAS, SQL, IBM SPSS**Software:** MS Office, Zotero, GPower, Tableau, REDCap**Research:** Systematic reviews, Real World Evidence, Health Economics, Pragmatic Trials, Target Trial Emulation**Languages:** Arabic (Native), English (Proficient), Spanish (Beginner)**PUBLICATIONS**

1. **Khalil, H.**, Guo, Yi, Bian, J., Guo, S., Yang, S., Lou, X., Scarton, L., Rogers, S.C., Wilkie, D.J., Dereemer, D., Allen, J, M, (In preparation). *Socioeconomic Disparities in the Use of Supportive Care Medications in Pancreatic Cancer Patients with Medicare*
2. **Khalil, H.**, Keshk, N., Khan, B., Gao, S., Khan, S., Campbell, N. L. (In preparation). *The Relationship Between Cumulative Anticholinergic Burden and Delirium Prevalence in Critically Ill Adults.*
3. Ahmed, I., Foster, D. R., Bushey, M. A., Campbell, N. L., **Khalil, H.**, Bobb, J. F., Justice, M., Kroenke, K., Albertson-Junkans, L., DeBar, L. L., Bradley, K. A. (Under review). *Concordance Between Electronic*

4. **Khalil, H.**, Keshk, N., Khan, B., Gao, S., Campbell, N. L. (2025). *Community-Based Anticholinergic Exposure and Delirium: A Systematic Review. Delirium*. Published online July 14, 2025. doi.org/10.56392/001c.141069
5. Noll, L. C., Holden, R. J., Gao, S., Unverzagt, F. W., Lane, K. A., Carter, A., Harrington, A. B., Manoharan, S., Manoharan, N., Rosenthal, D. L., Pitts, C., Pelkey, K., Papineau, E., Lauck, D. M., Keshk, N., Alamer, K., **Khalil, H.**, Boustani, M. A. (2024). *Deprescribing Anticholinergics to Preserve Brain Health (R2D2): Study Protocol for a Randomized Clinical Trial. Trials*, 25(1):788.
6. **Khalil, H.**, Lahoud, N. (2020). *Knowledge of Stroke Warning Signs, Risk Factors, and Response to Stroke Among Lebanese Older Adults in Beirut. Journal of Stroke & Cerebrovascular Diseases*, 29(5):104716.

PRESENTATIONS

1. **Khalil, H.**, Lahoud, N. (2019). *Knowledge of stroke warning signs, risk factors, and response to stroke among Lebanese older adults in Beirut*. Presented at the XIth International Epidemiological Association Eastern Mediterranean Regional Scientific Meeting, Beirut, Lebanon.
2. McDuffy, D., Alamer, K., **Khalil, H.**, Campbell, N. C. (2021). *Did I Do That? Comparing long-term cognitive effects of antimuscarinic overactive bladder medications in existing literature*. ASHP Midyear Clinical Meeting, Virtual.
3. Mize, A., Alamer, K., Ludlow, J., **Khalil, H.**, Campbell, N. (2021). *Symptom-guided recommendations to support clinical decision making in deprescribing anticholinergic overactive bladder medications*. ASHP Midyear Clinical Meeting, Virtual.
4. J. Ludlow, J., Mize, A., Alamer, K., **Khalil, H.**, Campbell, N. (2022). *Anticholinergic use and cognitive risk thresholds in primary care older adults*. American Geriatrics Society Annual Meeting, Orlando, FL.
5. **Khalil, H.**, Alamer, K., Campbell, N. (2022). *A systematic review of pre-hospital medication exposure on the risk of delirium*. American Delirium Society Annual Meeting, Indianapolis, IN.
6. **Khalil, H.**, Stachler, E., Alamer, K. A., Ludlow, J. A., Campbell, N. (2022). *Annualized and cumulative measures of anticholinergic exposure for research and clinical applications. Innovation in Aging*, 6 (Suppl 1), 622. PMID: PMC9766780.
7. Stachler, E., **Khalil, H.**, Alamer, K. A., Ludlow, J. A., Campbell, N. (2022). *Comparison of self-reported medication inventory and medication orders data among elderly anticholinergic users for quantification of medication risks*. ASHP Midyear Clinical Meeting, Virtual.
8. Keshk, N., **Khalil, H.**, Campbell, N. (2023). *Impact of anticholinergics on cognitive measures indicative of dementia: A comparative assessment of medication classes with anticholinergic activity*. Purdue College of Pharmacy Research Day, West Lafayette, IN.
9. Ahmed, I., **Khalil, H.**, Foster, D., Campbell, N. (2023). *Developing algorithms to reduce missingness of medication order data in electronic health records*. Purdue College of Pharmacy Research Day, West Lafayette, IN.
10. **Khalil, H.**, Campbell, N., Foster, D., Ahmed, I., Bushey, M. (2024). *Medication data in research: Improving completeness using rule-based algorithms on textual variables*. International Society for Pharmacoepidemiology (ISPE) Annual Meeting, Berlin, Germany. [966] <https://doi.org/10.1002/pds.5891>
11. Keshk, N., **Khalil, H.**, Campbell, N. (2024). *The relationship between anticholinergics and cognitive measures is important to the diagnosis of dementia. Alzheimer's & Dementia*, e095389. doi.org/10.1002/alz.095389
12. **Khalil, H.**, Campbell, N. (2024). *Association between cumulative anticholinergic burden and prevalent delirium in critically ill adults*. Presented at the Purdue College of Pharmacy Tyler Symposium, West Lafayette, IN.

13. Ahmed, I., Bushey, M., Campbell, N., **Khalil, H.**, Foster, D. (2024). *Outpatient pharmacological treatment characteristics in patients with comorbid opioid use disorder and depression*. Presented at the Purdue College of Pharmacy Tyler Symposium, West Lafayette, IN.
14. **Khalil, H.**, Al Habbab, T., Fowler, C. R., Nili, M., Keefe, R. S. (2024). *Investigational drugs for the treatment of cognitive impairments of schizophrenia: A review of trials*. Psych Congress 2024, Boston, MA.
15. Khalil, H., Al Habbab, T., Fowler, C. R., Nili, M., Keefe, R. S. (2024). *Investigational drugs for the treatment of cognitive impairments of schizophrenia: A review of trials*. Purdue College of Pharmacy Research Day
16. **Khalil, H.**, Khan, B., Khan, S., Boustani, M., Campbell, NC., (2025). *Association between cumulative anticholinergic burden and prevalent delirium in critically ill adults*. American Delirium Society Annual Meeting, Nashville, TN

PROFESSIONAL AFFILIATIONS

• Lebanese Order of Pharmacists	Since 2017
• International Society of Pharmacoeconomics and Outcomes Research	Since 2021
• American Delirium Society	Since 2021
• International Society of Pharmacoepidemiology	Since 2022
• American Association for Cancer Research	Since 2025